Full Text, Language, and Embeddings

Full Text, Language, and Embeddings I

Beyond bibliographic metadata, the pipeline mines the textual content of each record. This stage bridges the gap between a bibliographic inventory and a semantic understanding of the literature.

Full-Text Availability I

An open-access resolver maps each record to a downloadable PDF where one exists (a known `pdf_url`, or an Unpaywall lookup by DOI), and an opt-in, network-gated downloader fetches it to a deterministic path. Full-text availability is summarized without requiring any download, so the offline default still reports coverage. For this run:

Full-Text Availability I

- **Abstract coverage**: 61.6% of records (1437 of 2334) carry an abstract; 897 records lack one.
- **Open-access status**: 24.6% of records are open access (574 records); the remainder are closed or unknown.
- **PDF availability**: 54.6% of records (1275) have a direct PDF link; 1274 have a publisher PDF, and 1059 have no full-text source available.

Full-Text Availability I

The identifier coverage for this corpus is: 2260 DOIs, 923 OpenAlex IDs, and 1 arXiv IDs. DOI coverage dominates, enabling robust cross-engine de-duplication.

Language and Entity Extraction I

Titles, abstracts, and (when present) full text are tokenized and reduced to keyphrases and named entities by offline, dependency-light extractors — no mandatory LLM.

Term-frequency statistics drive a TF-IDF representation over a 500-feature vocabulary. The most frequent terms in the corpus are: modafinil, treatment, study, results, patients, effects, sleep, used, clinical, use, drug, studies, mg, using, sleepiness, disorder, associated, cognitive, narcolepsy, significant. These terms reflect the clinical, pharmacological, and cognitive vocabulary characteristic of the modafinil literature.

Every title, abstract, and full text is embedded into a shared vector space by a deterministic, offline method — TF-IDF followed by truncated SVD, i.e. latent semantic analysis (Deerwester et al. 1990). The embedding dimensionality is 50 components (configurable via `project_config.embeddings.n_components`), and the TF-IDF vocabulary is capped at 500 features (configurable via `project_config.embeddings.max_features`). The embedding is byte-stable across runs: the same input text always yields identical vectors, so the derived similarity matrix, nearest-neighbour lists, clusters, and two-dimensional projection are all reproducible.

An optional transformer backend can be enabled by setting `project_config.embeddings.method: transformer` (requires the `embeddings` extra), which upgrades the embedding fidelity without changing the interface or downstream analysis.

The embeddings support semantic retrieval over the corpus and feed two visualizations: a PCA two-dimensional projection ((Figure pca embeddings)) that maps the topical geography of the literature, and a hierarchical clustering dendrogram ((Figure dendrogram)) that reveals the similarity structure of the document collection.